

Agile climbers

Australia is home to some acrobatic marsupials that visit flowers in bushes and trees. This pygmy possum feeds on all kinds of food, including fruit, insects, and spiders, but it also drinks nectar from eucalyptus flowers. During the breeding season, females carry up to four young in their pouches as they climb high above the ground. A smaller marsupial, the honey possum from Western Australia relies entirely on nectar for its food.

Strong tail can provide extra grip by hanging onto stalks

Feet with opposable thumbs and nonslip pads on all five toes

Eucalyptus flower with no petals, but feathery stamens, which brush pollen onto animals

Ball of compressed pollen in basket of stiff hairs



Double diet

When honeybees visit flowers, they collect food in two ways. They drink nectar with their long tongues and at the same time they gather pollen on their body hairs. Every few seconds, a bee moistens its forelegs, then uses them to comb the pollen into baskets on its hind legs. When these are full, it carries the nectar and pollen back to the hive. Here, the nectar is used to make honey—an energy-rich food stored in honeycombs.

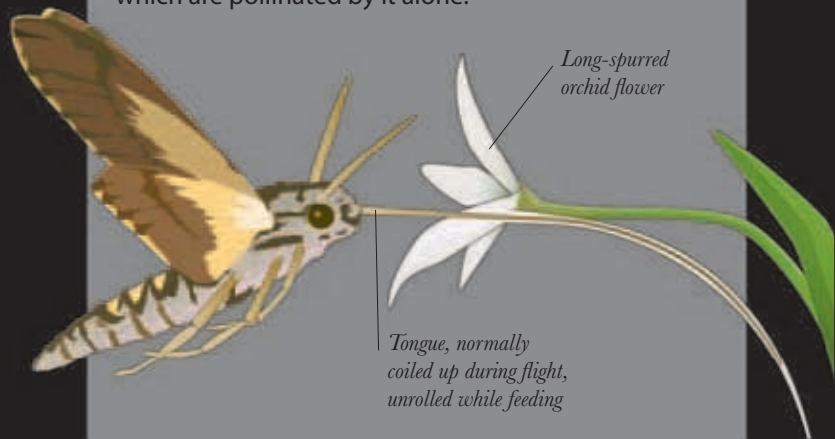
Pollen is used mainly to feed developing bee grubs.

Perfect fit

For more than 100 million years, flowering plants have evolved in step with their insect visitors. Some flowers are pollinated by a variety of insects, but the shape of others fits only one particular partner. Darwin's hawk moth of Madagascar has a tongue 12 in (30 cm) long. It visits flowers with extra-long nectar-tubes, or spurs, which are pollinated by it alone.

Long-spurred orchid flower

Tongue, normally coiled up during flight, unrolled while feeding



Protruding stamen brushes pollen onto bat's muzzle and tongue as it feeds

Big, broad wing creates a lot of lift, allowing bat to maneuver tightly around flower

